MTE544 Fall 2026 Assignment 2

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1 RANSAC with Ellipse I (18)

a.) The ellipse has five unknown coefficients: A, B, C, D, and E.

Each point $\mathbf{p}_i = (x_i, y_i)$ on the ellipse gives one independent linear equation in these unknowns. Therefore, the minimum number of points is **5**.

Moving the constant to the right side gives, for each point,

$$Ax_i^2 + 2Bx_iy_i + Cy_i^2 - 2Dx_i - 2Ey_i = -1.$$

Stacking the five equations gives the linear system $\mathbf{M}\boldsymbol{\theta} = -\mathbf{1}$:

$$\underbrace{\begin{bmatrix} x_1^2 & 2x_1y_1 & y_1^2 & -2x_1 & -2y_1 \\ x_2^2 & 2x_2y_2 & y_2^2 & -2x_2 & -2y_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_5^2 & 2x_5y_5 & y_5^2 & -2x_5 & -2y_5 \end{bmatrix}}_{\mathbf{M}\in\mathbb{R}^{5\times 5}} \underbrace{\begin{bmatrix} A \\ B \\ C \\ D \\ E \end{bmatrix}}_{\boldsymbol{\theta}} = \begin{bmatrix} -1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}.$$

If $\mathbf{M}$ is invertible, then solving for

$$\boldsymbol{\theta} = -\mathbf{M}^{-1}\mathbf{1}$$

will give the parameters A, B, C, D, and E from the given 5 points.

b.) Using the row pattern from part (a), the matrix $\mathbf{M}$ from the given points is

$$\mathbf{M} = \begin{bmatrix} 8.5264 & -35.0984 & 36.1201 & -5.84 & 12.02 \\ 11.5600 & -48.9600 & 51.8400 & -6.80 & 14.40 \\ 24.9001 & -78.2432 & 61.4656 & -9.98 & 15.68 \\ 30.0304 & -77.1584 & 49.5616 & -10.96 & 14.08 \\ 17.6400 & -49.6440 & 34.9281 & -8.40 & 11.82 \end{bmatrix}.$$

Solving $\mathbf{M}\boldsymbol{\theta} = -\mathbf{1}$ gives the polynomial coefficients

$$A = 0.018063, \quad B = 0.012069, \quad C = 0.030153, \quad D = -0.006250, \quad E = -0.154412.$$

The scale factor is

$$\alpha = \frac{1}{\begin{bmatrix} D & E \end{bmatrix} \begin{bmatrix} A & B \\ B & C \end{bmatrix}^{-1} \begin{bmatrix} D \\ E \end{bmatrix} - 1} = 41.678.$$

Then $\mathbf{Q}$ and $\mathbf{Qb}$ follow from

$$\mathbf{Q} = \alpha \begin{bmatrix} A & B \\ B & C \end{bmatrix} = \begin{bmatrix} 0.7528 & 0.5030 \\ 0.5030 & 1.2567 \end{bmatrix}, \qquad \mathbf{Qb} = \alpha \begin{bmatrix} D \\ E \end{bmatrix} = \begin{bmatrix} -0.2605 \\ -6.4356 \end{bmatrix}.$$

Solving for $\mathbf{b}$ gives

$$\mathbf{b} = \mathbf{Q}^{-1}(\mathbf{Qb}) = \begin{bmatrix} 4.199 \\ -6.802 \end{bmatrix}.$$

2 RANSAC with Ellipse II (30)

a.) The function *fit_ellipse_subset* is implemented in the submitted Python file.

b.) Five points are sampled at random without replacement, because five points are the minimum needed to determine the ellipse. The function *fit_ellipse_subset* solves for A through E and converts them to $\mathbf{Q}$ and $\mathbf{b}$. A sample is rejected if the fit fails or $\mathbf{Q}$ is not positive definite. A sample is also rejected if $\mathbf{b}$ lies outside the bounding box of the data, because the normalized residual test is scale dependent and accepts very large, near-degenerate ellipses. For each valid candidate, the residual $\left|\sqrt{(\mathbf{p}_i - \mathbf{b})^\top \mathbf{Q}(\mathbf{p}_i - \mathbf{b})} - 1\right|$ is computed for all data points. Points with a residual at most ε are inliers. The candidate with the most inliers is kept. This repeats for the requested number of iterations. The function returns $\mathbf{Q}$, $\mathbf{b}$, and the inliers.

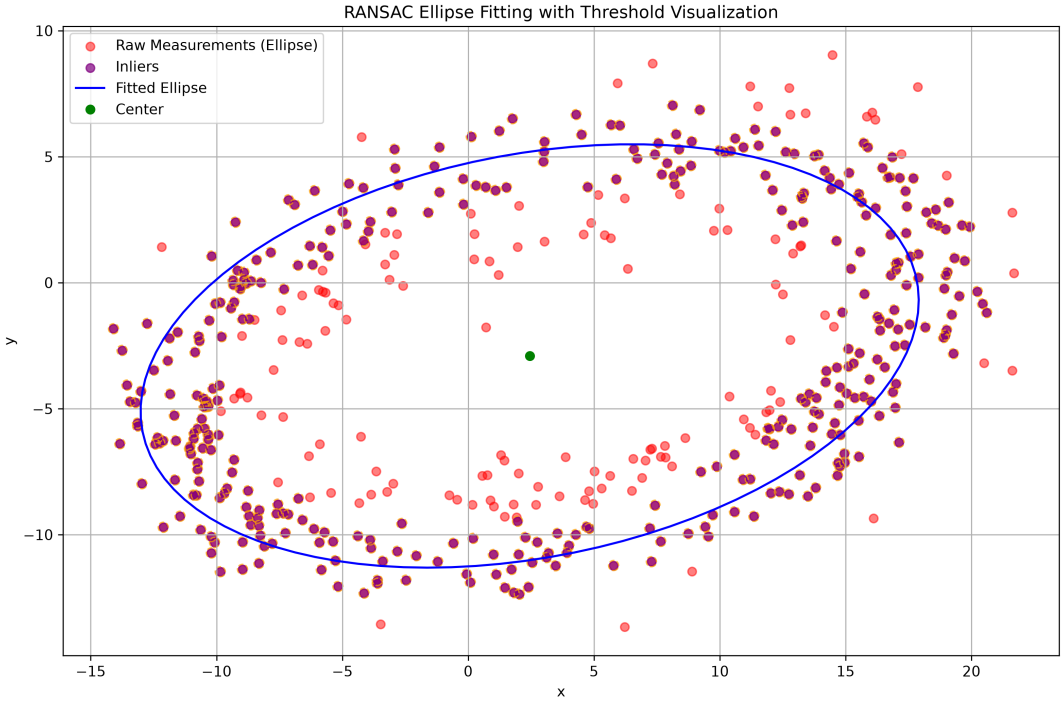


Figure 1: RANSAC ellipse fit with $N = 500$ points, 1000 iterations, and $\varepsilon = 0.2$. Inliers are shown in purple.

c.) Dataset sizes of 100, 500, and 1000 points kept the inlier fraction near 75%, so larger sets mainly add computation. More iterations give more chances to draw five inlier points. With about 75% inliers, roughly one sample in four is clean, so 100 iterations already suffice, and inliers rose only from 368 to 376 between 100 and 5000 iterations. Thresholds of 0.05, 0.2, and 0.5 gave 25%, 75%, and 99% inliers. A small threshold rejects valid noisy points, and a large threshold accepts outliers.

Dataset size	Iterations	Threshold	Inliers	Inlier fraction (%)
100	1000	0.20	75	75.0
500	1000	0.20	373	74.6
1000	1000	0.20	724	72.4
500	100	0.20	368	73.6
500	1000	0.20	373	74.6
500	5000	0.20	376	75.2
500	1000	0.05	127	25.4
500	1000	0.20	373	74.6
500	1000	0.50	496	99.2

Table 1: Inlier counts for different dataset sizes, iteration counts, and thresholds.

Note on the modified template

The template function *plot_ellipse* was modified. The original scaled the input matrix by 9.80665 and used its eigenvalues directly as axis lengths. This does not match the definition $(\mathbf{p}-\mathbf{b})^\top \mathbf{Q}(\mathbf{p}-\mathbf{b}) = 1$ used in this assignment. The function now uses semi-axis lengths of $1/\sqrt{\lambda_i}$ for the eigenvalues λ_i of $\mathbf{Q}$, and it requires $\mathbf{Q}$ to be positive definite.

3 Bayesian Filter for Tracking Moo-Deng's Behavior I (32)

a) By the Markov property, each move depends only on the current zone. The probability of the sequence is therefore the product of the four transition probabilities for $1 \to 2$, $2 \to 3$, $3 \to 3$, and $3 \to 1$.

$$P = A_{21}\, A_{32}\, A_{33}\, A_{13} = (0.4)(0.4)(0.7)(0.2) = 0.0224$$

The probability is 2.24%.

b) Since $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$, entry A_{ji} is the probability of moving from zone i to zone j. Each column holds the given probabilities for one starting zone, so each column sums to 1.

$$\mathbf{A} = \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.4 & 0.4 & 0.1 \\ 0 & 0.4 & 0.7 \end{bmatrix}$$

c) The stationary distribution satisfies $\mathbf{x}^* = \mathbf{A}\mathbf{x}^*$, which is $(\mathbf{A} - \mathbf{I})\mathbf{x}^* = \mathbf{0}$, together with $\mathbf{1}^\top \mathbf{x}^* = 1$. Writing $\mathbf{x}^* = [x_1,\ x_2,\ x_3]^\top$ gives

$$\begin{aligned} -0.4x_1 + 0.2x_2 + 0.2x_3 &= 0, \\ 0.4x_1 - 0.6x_2 + 0.1x_3 &= 0, \\ 0.4x_2 - 0.3x_3 &= 0. \end{aligned}$$

The three equations are linearly dependent, so the normalization $\mathbf{1}^\top \mathbf{x}^* = 1$ is needed to obtain a unique solution.

The third equation gives

$$x_3 = \tfrac{4}{3} x_2.$$

Substituting into the first equation gives

$$-0.4x_1 + 0.2x_2 + 0.2\left(\tfrac{4}{3}\right) x_2 = 0 \;\Rightarrow\; x_1 = \tfrac{7}{6} x_2.$$

The normalization gives

$$\left(\tfrac{7}{6} + 1 + \tfrac{4}{3}\right) x_2 = \tfrac{7}{2} x_2 = 1 \;\Rightarrow\; x_2 = \tfrac{2}{7}.$$

Back-substitution gives

$$\mathbf{x}^* = \begin{bmatrix} 1/3 \\ 2/7 \\ 8/21 \end{bmatrix} \approx \begin{bmatrix} 0.333 \\ 0.286 \\ 0.381 \end{bmatrix}.$$

The unused second equation checks the result: $0.4\left(\tfrac{1}{3}\right) - 0.6\left(\tfrac{2}{7}\right) + 0.1\left(\tfrac{8}{21}\right) = 0$.

d) Since the sensor probabilities equal $\mathbf{C}\mathbf{x}_k$, entry C_{ji} is the probability that the sensor reports zone j when Moo-Deng is in zone i. Each column holds the accuracy and error rates for one true zone, so each column sums to 1.

$$\mathbf{C} = \begin{bmatrix} 0.8 & 0.2 & 0.05 \\ 0.1 & 0.7 & 0.1 \\ 0.1 & 0.1 & 0.85 \end{bmatrix}$$

e) The filter assumes that the motion follows the Markov model and that each measurement depends only on the current zone. Let $\mathbf{x}_{k-1}$ be the belief after the previous measurement, so $x_{k-1}[j] = p(A_{k-1} = j \mid \mathbf{y}_{1:k-1})$.

Prediction. By the law of total probability,

$$\bar{x}_k[i] = p(A_k = i \mid \mathbf{y}_{1:k-1}) = \sum_{j=1}^{3} p(A_k = i \mid A_{k-1} = j)\, x_{k-1}[j], \qquad \bar{\mathbf{x}}_k = \mathbf{A}\mathbf{x}_{k-1}.$$

Update. By Bayes' rule,

$$p(A_k = i \mid \mathbf{y}_{1:k}) = \frac{p(\mathbf{y}_k \mid A_k = i)\, \bar{x}_k[i]}{p(\mathbf{y}_k \mid \mathbf{y}_{1:k-1})}.$$

If the sensor reports $\mathbf{y}_k = \mathbf{e}_m$, the likelihood is $p(\mathbf{y}_k \mid A_k = i) = C_{mi}$. This is entry i of $\mathbf{C}^\top \mathbf{y}_k$, which selects row m of $\mathbf{C}$.

Combined estimator. The denominator is a constant that makes the belief sum to 1. Therefore,

$$\mathbf{x}_k = \frac{\operatorname{diag}\big(\mathbf{C}^\top \mathbf{y}_k\big)\, \mathbf{A}\mathbf{x}_{k-1}}{\mathbf{1}^\top \operatorname{diag}(\mathbf{C}^\top \mathbf{y}_k)\, \mathbf{A}\mathbf{x}_{k-1}} = \frac{\operatorname{diag}\big(\mathbf{C}^\top \mathbf{y}_k\big)\, \mathbf{A}\mathbf{x}_{k-1}}{\mathbf{y}_k^\top \mathbf{C}\mathbf{A}\mathbf{x}_{k-1}}.$$

The estimated zone is

$$\hat{A}_k = \arg\max_i \, x_k[i].$$

For example, if $\mathbf{x}_{k-1} = [1,\ 0,\ 0]^\top$ and the sensor reports the Resting Area, then $\mathbf{A}\mathbf{x}_{k-1} = [0.6,\ 0.4,\ 0]^\top$ and the likelihood is $[0.1,\ 0.7,\ 0.1]^\top$. The normalized product is $\mathbf{x}_k \approx [0.176,\ 0.824,\ 0]^\top$.

4 Bayesian Filter for Tracking Moo-Deng's Behavior II (20)

a) *moodeng_behavior_update* uses the column of $\mathbf{A}$ for the current state as probabilities and samples the next state. Repeating this from each state many times, the empirical transition frequencies should approach that column. Long-run zone frequencies should also approach the stationary distribution $\mathbf{x}^*$.

b) *sensor_measurement* uses the column of $\mathbf{C}$ for the true zone as probabilities and samples the reported zone. The function returns the matching one-hot vector, $[1,0,0]^\top$, $[0,1,0]^\top$, or $[0,0,1]^\top$.

c) The function *sim_moodeng* first sets $\mathbf{A}$ and $\mathbf{C}$ to the values found in Problem 3. The initial belief $\mathbf{x}_0$ places probability 1 on the known initial zone. At each step, the simulation samples the true next zone and the sensor reading. The estimator then implements the Bayes filter derived in Problem 3(e):

$$\bar{\mathbf{x}}_k = \mathbf{A}\mathbf{x}_{k-1}, \qquad \mathbf{x}_k = \frac{\operatorname{diag}\big(\mathbf{C}^\top \mathbf{y}_k\big)\,\bar{\mathbf{x}}_k}{\mathbf{1}^\top \operatorname{diag}(\mathbf{C}^\top \mathbf{y}_k)\,\bar{\mathbf{x}}_k}, \qquad \hat{A}_k = \arg\max_i x_k[i].$$

The belief $\mathbf{x}_k$ then becomes the prior for the next step.

Table 2 shows one simulation of 20 steps starting in the Feeding Area ($A_0 = 1$). Zones are numbered 1 for Feeding, 2 for Resting, and 3 for Playground. The first Playground probability is exactly zero, because Moo-Deng starts in the Feeding Area with certainty and cannot move directly to the Playground Area. The estimated zone matched the sensor reading at every step of this run. It differed from the true zone at five of the 20 steps (steps 4, 5, 6, 10, and 13), where the sensor itself reported the wrong zone.

k	True A_k	Sensor	Estimated $\hat{A}_k$	$x_k[1]$	$x_k[2]$	$x_k[3]$
1	2	2	2	0.176	0.824	0.000
2	2	2	2	0.080	0.824	0.097
3	3	3	3	0.058	0.093	0.849
4	3	1	1	0.747	0.122	0.132
5	1	2	2	0.158	0.798	0.045
6	1	2	2	0.079	0.815	0.106
7	2	2	2	0.072	0.803	0.125
8	3	3	3	0.056	0.089	0.854
9	2	2	2	0.120	0.540	0.340
10	2	1	1	0.707	0.212	0.081
11	2	2	2	0.148	0.808	0.044
12	2	2	2	0.078	0.815	0.106
13	3	1	1	0.664	0.264	0.072
14	3	3	3	0.215	0.174	0.611
15	1	1	1	0.770	0.146	0.084
16	2	2	2	0.156	0.808	0.036
17	2	2	2	0.079	0.817	0.104
18	1	1	1	0.664	0.264	0.072
19	2	2	2	0.142	0.810	0.048
20	1	1	1	0.684	0.257	0.059

Table 2: Simulation of Moo-Deng's behavior with $A_0 = 1$. The columns $x_k[1]$, $x_k[2]$, and $x_k[3]$ are the belief probabilities of the Feeding, Resting, and Playground Areas.